NETWORKING / NET02 / NOVEMBER 2026

Tracing & Containing Suspicious Network Endpoints

Find the Imposter - Layer 2

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The task

I started with three flagged MAC addresses and an unfamiliar switched network. My task was to locate each host from Switch0, identify its attachment port, and take it off the network without disconnecting other users.

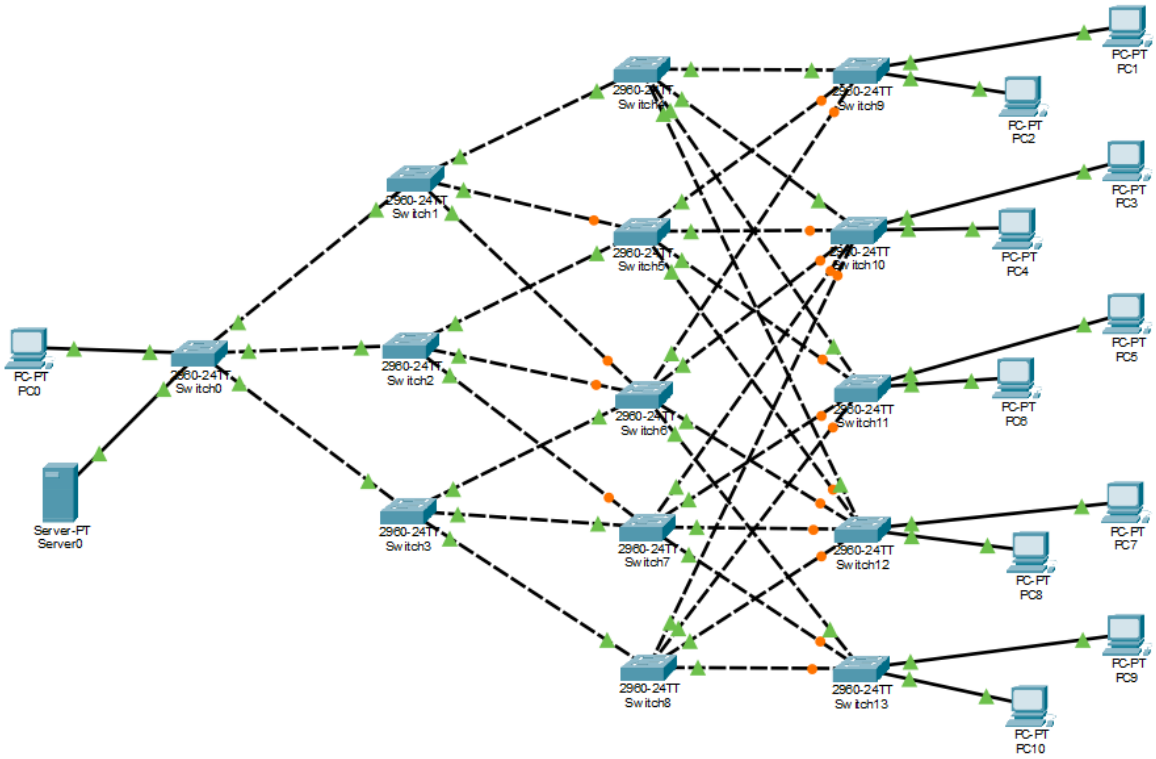


Figure 1. The supplied redundant Layer 2 topology. I followed MAC learning and CDP rather than guessing a route from the drawing.

3 targets	3 endpoint ports	8 remaining responders
All traced to attachment ports	Disabled and saved	Observed after containment

Lab credit and scope

The topology and challenge are from Practical Networking, "Find the Imposter - Layer 2." This case study records my investigation with coaching on command use and evidence interpretation. It is a simulation, not a production incident. The malicious designation comes from the lab brief; I did not independently detect or analyze malicious traffic.

01 / METHOD & INITIAL OBSERVATION

Make the evidence usable before following it

The challenge restricted access to PC1 through PC10 and required an evidence-based path from Switch0. I used switch CLI output and cable inspection, without opening those endpoint consoles.

My investigation sequence

1. Generate traffic so the switches can learn source MAC addresses.
2. Find each target in the current MAC table.
3. Match the learned port to its CDP neighbor.
4. Repeat on that neighbor until reaching an endpoint attachment.
5. Confirm the port, contain the host, then check the remaining network.

At first, Switch0 showed only three non-target MAC entries. I had stopped PC0's continuous broadcast ping. An incomplete table was not enough to conclude that a target was absent. Aging was a plausible explanation, but I did not measure the aging timer.

```
Switch0#show mac addr
Switch0#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    000c.85a5.a401    DYNAMIC     Fa0/1
   1    0030.f2ed.c701    DYNAMIC     Fa0/4
   1    00d0.baa9.9301    DYNAMIC     Fa0/5
Switch0#
```

Figure 2. The first lookup did not contain any of the three target MAC addresses.

I restarted ping -t 10.9.9.255 on PC0 and left it running. Replies from the lab hosts generated return frames, and the refreshed table contained the targets. This corrected the evidence gap without changing switch configuration.

```
Switch0#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0000.0c07.be8e    DYNAMIC     Fa0/1
   1    0001.9666.3d1b    DYNAMIC     Fa0/1
   1    0001.c92c.7351    DYNAMIC     Fa0/3
   1    0005.5ea7.bb7a    DYNAMIC     Fa0/1
   1    0006.2a4d.6e7a    DYNAMIC     Fa0/2
   1    0006.2a55.34de    DYNAMIC     Fa0/1
   1    0007.ec7a.b41a    DYNAMIC     Fa0/1
   1    000c.85a5.a401    DYNAMIC     Fa0/1
   1    0010.1154.7611    DYNAMIC     Fa0/5
   1    0030.a345.d1bb    DYNAMIC     Fa0/1
   1    0030.f2ed.c701    DYNAMIC     Fa0/4
   1    00d0.973b.d35c    DYNAMIC     Fa0/1
   1    00d0.baa9.9301    DYNAMIC     Fa0/5
```

Figure 3. After traffic resumed, all three target entries were visible on Switch0 Fa0/1.

02 / FOLLOW THE NEXT HOP

A learned port gives direction, not the final location

On Switch0, all three targets pointed to Fa0/1. CDP identified Switch1 on that interface. I treated it as the next hop, not a port to shut down: it carried multiple hosts beyond the local switch.

```
Switch0#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0000.0c07.be8e    DYNAMIC     Fa0/1
   1    0001.9666.3d1b    DYNAMIC     Fa0/1
   1    0001.c92c.7351    DYNAMIC     Fa0/3
   1    0005.5ea7.bb7a    DYNAMIC     Fa0/1
   1    0006.2a4d.6e7a    DYNAMIC     Fa0/2
   1    0006.2a55.34de    DYNAMIC     Fa0/1
   1    0007.ec7a.b41a    DYNAMIC     Fa0/1
   1    000c.85a5.a401    DYNAMIC     Fa0/1
   1    0010.1154.7611    DYNAMIC     Fa0/5
   1    0030.a345.d1bb    DYNAMIC     Fa0/1
   1    0030.f2ed.c701    DYNAMIC     Fa0/4
   1    00d0.973b.d35c    DYNAMIC     Fa0/1
   1    00d0.baa9.9301    DYNAMIC     Fa0/5
   1    00d0.d372.7800    DYNAMIC     Fa0/1
   1    00e0.f92d.29a7    DYNAMIC     Fa0/5
Switch0#
Switch0#
Switch0#show cdp neh
Switch0#show cdp ne
Switch0#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID     Local Intrfce   Holdtme    Capability   Platform    Port ID
Switch1       Fas 0/1          168            S       2960        Fas 0/1
Switch2       Fas 0/4          168            S       2960        Fas 0/1
Switch3       Fas 0/5          168            S       2960        Fas 0/1
Switch0#
```

Figure 4. Switch0: target entries on Fa0/1; CDP maps local Fa0/1 to Switch1.

Switch1 learned the same targets through Fa0/2. Its CDP table identified Switch4 on that local interface. The shared path was now Switch0 to Switch1 to Switch4.

```
Switch1#show mac address-table ad
Switch1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0000.0c07.be8e    DYNAMIC     Fa0/2
   1    0001.9666.3d1b    DYNAMIC     Fa0/2
   1    0005.5ea7.bb7a    DYNAMIC     Fa0/2
   1    0006.2a4d.6e7a    DYNAMIC     Fa0/1
   1    0006.2a55.34de    DYNAMIC     Fa0/2
   1    0007.ec17.4b01    DYNAMIC     Fa0/2
   1    0007.ec7a.b41a    DYNAMIC     Fa0/2
   1    000b.be68.9801    DYNAMIC     Fa0/1
   1    0030.a345.d1bb    DYNAMIC     Fa0/2
   1    00d0.973b.d35c    DYNAMIC     Fa0/2
   1    00d0.978d.9b01    DYNAMIC     Fa0/3
   1    00d0.d372.7800    DYNAMIC     Fa0/2
   1    00e0.a36b.6c01    DYNAMIC     Fa0/4
Switch1#show cd
Switch1#show cdp n
Switch1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID     Local Intrfce   Holdtme    Capability   Platform    Port ID
Switch0       Fas 0/1          141            S       2960        Fas 0/1
Switch5       Fas 0/3          141            S       2960        Fas 0/1
Switch6       Fas 0/4          141            S       2960        Fas 0/1
Switch4       Fas 0/2          141            S       2960        Fas 0/1
Switch1#
```

Figure 5. Switch1: target entries on Fa0/2; CDP maps that interface to Switch4.

A dynamic MAC entry records the port on which a switch received frames with that source MAC. CDP supplies neighbor identity and local/remote interface mapping. Using the two together kept every hop tied to observable evidence.

03 / NARROW THE SEARCH

The three paths split at Switch4

At Switch4, the targets were learned through three different interfaces. I matched each interface to a CDP neighbor, then followed one target to completion before returning to the others.

```
Switch4#show mac a
Switch4#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0000.0c07.be8e    DYNAMIC     Fa0/3
   1    0001.9644.2101    DYNAMIC     Fa0/3
   1    0001.9666.3d1b    DYNAMIC     Fa0/2
   1    0002.173d.0701    DYNAMIC     Fa0/2
   1    0005.5ea7.bb7a    DYNAMIC     Fa0/4
   1    0006.2a4d.6e7a    DYNAMIC     Fa0/1
   1    0006.2a55.34de    DYNAMIC     Fa0/5
   1    0007.ec7a.b41a    DYNAMIC     Fa0/2
   1    000c.85a5.a402    DYNAMIC     Fa0/1
   1    0030.a345.d1bb    DYNAMIC     Fa0/5
   1    0060.708b.a501    DYNAMIC     Fa0/4
   1    00d0.973b.d35c    DYNAMIC     Fa0/3
   1    00d0.d372.7800    DYNAMIC     Fa0/4
   1    00e0.8f52.b701    DYNAMIC     Fa0/5
Switch4#show cd
Switch4#show cdp n
Switch4#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID    Local Intrfce   Holdtme    Capability   Platform    Port ID
Switch9      Fas 0/2          124            S       2960        Fas 0/1
Switch11     Fas 0/4          124            S       2960        Fas 0/1
Switch1      Fas 0/1          124            S       2960        Fas 0/2
Switch10     Fas 0/3          124            S       2960        Fas 0/1
Switch12     Fas 0/5          124            S       2960        Fas 0/1
Switch4#
```

Figure 6. Switch4 provides the branch point: MAC-table ports matched to CDP neighbor identities.

Target MAC	Switch4 local port	Next switch / remote port
0001.9666.3d1b	Fa0/2	Switch9 / Fa0/1
0000.0c07.be8e	Fa0/3	Switch10 / Fa0/1
0006.2a55.34de	Fa0/5	Switch12 / Fa0/1

Separate uplinks from attachment ports

These Switch4 interfaces were transit links. Disabling one here could disconnect an entire downstream branch. I continued to the next switch to locate the actual host attachment.

The path reflected the learned forwarding state at the time of inspection. The redundant links in the topology were not an instruction to visit every switch, and this exercise did not include a separate spanning-tree or failover investigation.

04 / FIRST ENDPOINT

No CDP neighbor does not mean no device

Switch9 learned 0001.9666.3d1b on Fa0/4. Its CDP table listed switch neighbors on Fa0/1, Fa0/2, and Fa0/3, with no neighbor on Fa0/4. That suggested an endpoint-facing port, but CDP absence alone was not proof.

```
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0001.9666.3d1b    DYNAMIC     Fa0/4
   1    0006.2a4d.6e7a    DYNAMIC     Fa0/1
   1    0007.ec17.4b02    DYNAMIC     Fa0/1
   1    0007.ec7a.b41a    DYNAMIC     Fa0/5
Switch9#show cdp
Switch9#show cdp ne
Switch9#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID    Local Intrfce   Holdtme    Capability   Platform    Port ID
Switch5      Fas 0/2          123            S       2960        Fas 0/3
Switch6      Fas 0/3          123            S       2960        Fas 0/4
Switch4      Fas 0/1          123            S       2960        Fas 0/2
Switch9#
```

Figure 7. Switch9 associates the first target with Fa0/4. The CDP neighbor list has no Fa0/4 entry.

I checked link status and switchport state. Fa0/4 was connected and operating as static access in VLAN 1. Its administrative mode was dynamic auto; I did not mistake the operational access state for an explicitly configured switchport mode access command.

```
Switch9#show interfaces FastEthernet0/4 stat
Switch9#show interfaces FastEthernet0/4 status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/4                        connected    1          a-full  a-100 10/100BaseTX

Switch9#show interfaces FastEthernet0/4 swi
Switch9#show interfaces FastEthernet0/4 switchport
Name: Fa0/4
Switchport: Enabled
Administrative Mode: dynamic auto
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
 --More--
```

Figure 8. Fa0/4 is connected, with operational access mode and access VLAN 1.

Cable inspection then confirmed Switch9 Fa0/4 connected to PC1. This combined the learned MAC, interface state, neighbor evidence, and physical attachment before containment. PCs do not normally advertise CDP neighbors.

05 / COMPLETE THE TRACE

Apply the same checks to the other two targets

I returned to the two remaining branches from Switch4 and repeated the same lookup. Both target ports were absent from the respective CDP neighbor lists. Separate status and switchport checks showed them connected and operating as access ports in VLAN 1.

```
Switch10#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0000.0c07.be8e    DYNAMIC     Fa0/7
   1    0006.2a4d.6e7a    DYNAMIC     Fa0/1
   1    0007.ec17.4b03    DYNAMIC     Fa0/1
   1    00d0.973b.d35c    DYNAMIC     Fa0/6
Switch10#show cdp ne
Switch10#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID    Local Intrfce   Holdtme    Capability   Platform    Port ID
Switch8      Fas 0/3          136            S       2960        Fas 0/5
Switch7      Fas 0/4          136            S       2960        Fas 0/6
Switch5      Fas 0/2          136            S       2960        Fas 0/4
Switch6      Fas 0/5          136            S       2960        Fas 0/5
Switch4      Fas 0/1          136            S       2960        Fas 0/3
Switch10#
```

Figure 9. Switch10: 0000.0c07.be8e is learned on Fa0/7; CDP lists the switch-facing interfaces.

```
Switch12#show mac ad
Switch12#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0006.2a4d.6e7a    DYNAMIC     Fa0/1
   1    0006.2a55.34de    DYNAMIC     Fa0/7
   1    0007.ec17.4b05    DYNAMIC     Fa0/1
   1    0030.a345.d1bb    DYNAMIC     Fa0/6
Switch12#show dc
Switch12#show dcp ne
Switch12#show cdp ne
Switch12#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID    Local Intrfce   Holdtme    Capability   Platform    Port ID
Switch8      Fas 0/3          138            S       2960        Fas 0/3
Switch7      Fas 0/4          138            S       2960        Fas 0/4
Switch5      Fas 0/2          138            S       2960        Fas 0/6
Switch6      Fas 0/5          138            S       2960        Fas 0/6
Switch4      Fas 0/1          138            S       2960        Fas 0/5
Switch12#
```

Figure 10. Switch12: 0006.2a55.34de is learned on Fa0/7; CDP again identifies the transit interfaces.

MAC	Final attachment	Endpoint
0001.9666.3d1b	Switch9 Fa0/4	PC1
0000.0c07.be8e	Switch10 Fa0/7	PC4
0006.2a55.34de	Switch12 Fa0/7	PC8

I confirmed the PC labels by inspecting their cable connections. The post-containment topology captures all three attachments together. The completed result depends on the switch and port mapping, not on guessing an IP address from a PC name.

06 / CONTAIN & SAVE

Disable only the confirmed attachment ports

I used shutdown on Switch9 Fa0/4, Switch10 Fa0/7, and Switch12 Fa0/7. Each interface then reported disabled. I copied the running configuration to startup configuration on all three switches; each save returned [OK].

```
Switch9#
%SYS-5-CONFIG_I: Configured from console by console

Switch9#show interfaces FastEthernet0/4 status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/4                        disabled 1          auto    auto  10/100BaseTX

Switch9#
Switch9#
Switch9#copy run
Switch9#copy running-config str
Switch9#copy running-config sta
Switch9#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Switch9#
```

Figure 11. Switch9 Fa0/4: disabled status and successful configuration save.

```
Switch10(config-if)#end
Switch10#
%SYS-5-CONFIG_I: Configured from console by console

Switch10#show interfaces f
Switch10#show interfaces fastEthernet0/7 status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/7                        disabled 1          auto    auto  10/100BaseTX

Switch10#copy con
Switch10#copy run
Switch10#copy running-config st
Switch10#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Switch10#
```

Figure 12. Switch10 Fa0/7: disabled status and successful configuration save.

```
Switch12(config-if)#end
Switch12#
%SYS-5-CONFIG_I: Configured from console by console

Switch12#show interfaces fastEthernet0/7 status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/7                        disabled 1          auto    auto  10/100BaseTX

Switch12#copy run
Switch12#copy running-config st
Switch12#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Switch12#
```

Figure 13. Switch12 Fa0/7: disabled status and successful configuration save.

The saves are directly evidenced. I did not reload the switches to test persistence across a restart. A MAC entry disappearing by itself would be weaker proof because dynamic entries can age out; the disabled interface state verifies the administrative action.

07 / VALIDATE THE IMPACT

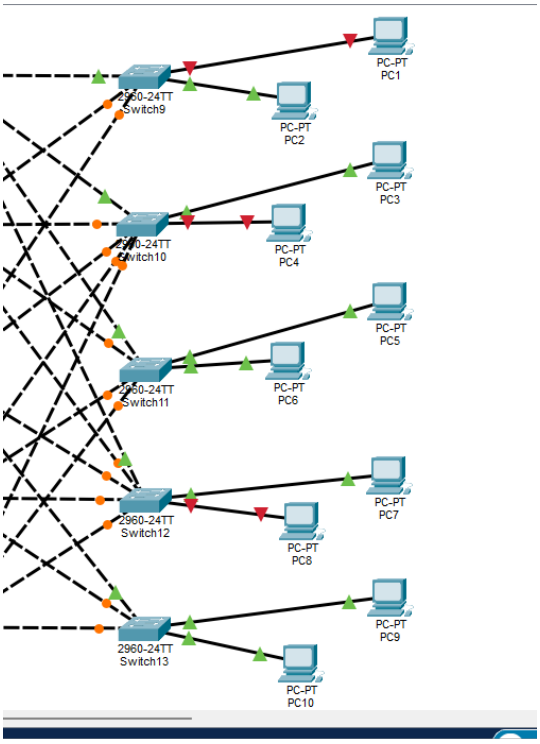
Check both the targets and the remaining network

On PC0, I stopped the earlier ping and started a fresh broadcast ping to 10.9.9.255. The captured output contains eight distinct responding IP addresses, compared with eleven in the initial test. The endpoint topology shows red links only for the three selected PCs; the other endpoint links remain green.

```
C:\>ping -t 10.9.9.255

Pinging 10.9.9.255 with 32 bytes of data:

Reply from 10.9.9.254: bytes=32 time<1ms TTL=128
Reply from 10.9.9.105: bytes=32 time<1ms TTL=128
Reply from 10.9.9.110: bytes=32 time<1ms TTL=128
Reply from 10.9.9.103: bytes=32 time<1ms TTL=128
Reply from 10.9.9.108: bytes=32 time<1ms TTL=128
Reply from 10.9.9.106: bytes=32 time<1ms TTL=128
Reply from 10.9.9.109: bytes=32 time<1ms TTL=128
Reply from 10.9.9.102: bytes=32 time<1ms TTL=128
Reply from 10.9.9.254: bytes=32 time<1ms TTL=128
Reply from 10.9.9.105: bytes=32 time<1ms TTL=128
Reply from 10.9.9.103: bytes=32 time<1ms TTL=128
Reply from 10.9.9.106: bytes=32 time<1ms TTL=128
Reply from 10.9.9.109: bytes=32 time<1ms TTL=128
Reply from 10.9.9.102: bytes=32 time<1ms TTL=128
Reply from 10.9.9.110: bytes=32 time=1ms TTL=128
Reply from 10.9.9.108: bytes=32 time=1ms TTL=128
Reply from 10.9.9.254: bytes=32 time<1ms TTL=128
Reply from 10.9.9.105: bytes=32 time<1ms TTL=128
Reply from 10.9.9.103: bytes=32 time<1ms TTL=128
Reply from 10.9.9.106: bytes=32 time<1ms TTL=128
Reply from 10.9.9.109: bytes=32 time<1ms TTL=128
Reply from 10.9.9.102: bytes=32 time<1ms TTL=128
Reply from 10.9.9.110: bytes=32 time=11ms TTL=128
Reply from 10.9.9.108: bytes=32 time=11ms TTL=128
```



Figures 14-15. Fresh PC0 responses and endpoint links after all three shutdowns. PC1, PC4, and PC8 are disconnected; the other eight destinations still reply.

Observed after containment	Addresses
Eight distinct ICMP responders	10.9.9.254, 10.9.9.102, 10.9.9.103, 10.9.9.105, 10.9.9.106, 10.9.9.108, 10.9.9.109, 10.9.9.110
No longer present in the captured reply set	10.9.9.101, 10.9.9.104, 10.9.9.107

What this verifies

The eight replies demonstrate continued ICMP reachability from PC0 to those destinations during the test. The target-port shutdowns and disconnected links provide the containment evidence. This was not an application-service test or an exhaustive all-pairs connectivity check.

Broadcast ping receives multiple replies for one request in this simulator. Its aggregate sent/received counters are not a normal one-to-one packet-loss measurement; I compared distinct responders instead.

08 / HANDOFF & LESSONS

A short, repeatable investigation

The result

I located all three flagged MAC addresses without inspecting the restricted PCs. The learned path led through Switch1 and Switch4, then to the correct attachment on Switch9, Switch10, or Switch12. I disabled those three ports, saved the configurations, and confirmed that eight other destinations continued replying.

Commands and their purpose

Command	Why I used it
ping -t 10.9.9.255 (PC0)	Keep return traffic flowing in this lab so dynamic MAC learning stays useful.
show mac address-table	Find the current learned interface for each target MAC.
show cdp neighbors	Map that local interface to the next switch and remote port.
show interfaces FastEthernet0/x status	Check connected state before the change and disabled state afterward.
show interfaces FastEthernet0/x switchport	Check operational mode and access VLAN.
configure terminal; interface FastEthernet0/x; shutdown; end	Select and administratively disable the confirmed endpoint port. Enter as separate commands.
copy running-config startup-config	Save the current configuration. Each of the three saves returned [OK].

What I would carry into a support role

Start with current evidence, narrow the path one hop at a time, and distinguish a host attachment from a shared uplink. Record the switch, port, VLAN, and endpoint before making a change. Verify the intended action and check what still works afterward. These habits support LAN troubleshooting, device location, and incident containment.

Boundaries

The exercise supplied the flagged MACs. It does not prove malware detection, attacker attribution, or host remediation. I used coaching while interpreting the outputs, and the topology was provided by the author. No reload test was performed. I have not included the author's Packet Tracer file for redistribution; this publication contains my write-up and selected screenshots.

Source: Practical Networking - Packet Tracer labs, Find the Imposter - Layer 2
Portfolio: carlostibj.github.io/project-net02.html
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